# CFOP · the easy way

yellow on top · white cross on the bottom · green in front

**sexy** `R U R' U'` **sledgehammer** `R' F R F'` **fat f** `f (front two layers)` **fat sexy** `r U R' U'`

## 1 · Cross

White cross on the bottom, edges matching their centres. Basic method; advanced cross planning comes later.



1. **Daisy.** Bring the 4 white edges up around the yellow centre, white facing up. Turn U out of the way before moving an edge up so you do not knock one off.
2. **Match and drop.** Turn U until a daisy edge's side colour matches the centre below it, then turn that face twice (F2 / R2 / B2 / L2). Repeat for all 4.
3. **Next step up: direct cross.** Skip the daisy: build the white cross straight on the bottom, each edge matched to its centre as you place it. Fewer moves, closer to real CFOP.
4. **Check.** Flip to look at the bottom: a white plus, and each edge's side colour matches its centre.

## 2 · F2L

Pair each white corner with its edge in the top layer, then insert the pair. Four pairs. Cross stays on the bottom. From J Perm.

1. **Order.** Both pieces into the top layer, look where the white sticker faces, pair them, insert. Never break a slot you have already solved. Examples go into the front-right slot; the left side is the mirror.
2. **Finger rule.** Index finger on top, parallel to the white sticker: the two edges it touches are where the edge can go (together or apart).



### Insert right, joined pair

Joined pair at the front-left of the top, white facing left; goes into the front-right slot

**`R U' R'`**

slot up · pair over · slot down



### Insert left, joined pair

Joined pair at the front-right of the top, white facing right; goes into the front-left slot

**`L' U L`**

mirror of the right-hand insert



### Insert right, split pair

Corner above the front-right slot, white facing right; edge at the back; top colours don't match

**`R U R'`**

slot up · corner over · slot down



### Insert left, split pair

Corner above the front-left slot, white facing left; edge at the back; top colours don't match

**`L' U' L`**

mirror of the right-hand insert



### Piece stuck in a slot

Corner in its slot, white facing front; edge on the right of the top

Lift it out with an insert move. Here that leaves a joined pair: U, insert.

**`R U' R' · U · R U' R'`**



### White on the side, colours match

Corner over its slot, white facing front; edge at the back; top colours match

U' (white still showing). Match, so together: R away, U edge next to it, R' back. U2, insert.

**`U' · R U R' · U2 · R U' R'`**



### White on the side, colours don't match

Corner over its slot, white facing right; edge on the left; top colours don't match

U (white still showing). No match, so apart: F' away, U edge to the far side, F back. U', insert.

**`U · F' U F · U' · R U R'`**



### White facing up

Corner over its slot, white facing up; edge at the back

U until the edge matches its centre. R away (edge to the back), U2 corner on top of it, R' back. U, insert.

**`U · R U2 R' · U · R U' R'`**



### Special case: split first

Corner over its slot, white facing up; edge next to it on the right

The white-up rule goes round in a circle. Split with R U2 R' (now apart), U', insert.

**`R U2 R' · U' · R U R'`**

## 3 · OLL step 1 · yellow cross

edges only; corners ignored (grey)

Dot (no yellow edges): do Minus, you get an L, then do L.



### Minus

Line left to right

**`F R U R' U' F'`**

F · sexy · F'



### L

L on the front and right

**`f R U R' U' f'`**

fat f · sexy · fat f'

## 4 · OLL step 2 · yellow corners

yellow cross already on top; only yellow shown



### Plus (double headlight)

Headlights front and back

**(R U2 R' U' R U' R') ×2**

Antisune, then Antisune

or F (R U R' U')3 F'



### Sune (fish) standard

Yellow corner front-left; yellow on the front face, right corner

**R U R' U R U2 R'**

take the front-right pair out, round, back in



### Antisune (fish)

standard

Yellow corner back-right; yellow on the front face, left corner

**R U2 R' U' R U' R'**

out with U2, over, back in



### Plus (headlight and blink)

Headlights on the left, blinker facing you

**R U2 R2 U' R2 U' R2 U2 R**

R U2 out · R2 U' out · R2 U' in · R2 U2 R in



### Bowtie

Solved corner front-right; x to face the white

**x R' U R D' R' U' R D x'**

x · R' U R D' · R' U' R D · x'



### Chameleon (headlight)

Headlights facing you

**R2 D R' U2 R D' R' U2 R'**

R2 D · R' U2 R out · D' · R' U2 R'



### Chameleon (blink)

Blink on the left

**r U R' U' r' F R F'**

fat sexy out · sledgehammer in

## 5 · PLL step 1 · corners

look for headlights (two same-colour corner stickers on one side)



### Adjacent corner swap

Solved corners on the left, swap the right two

**R U R' U' R' F R2 U' R' U' R U R' F'**

sexy · R' F · R2 U' R' U' · R U R' F'



### Diagonal corner swap (Y perm)

No headlights anywhere; any angle

**F R U' R' U' R U R' F' R U R' U' R' F R F'**

F · inverse sexy · 3-move sexy · undo F · sexy · sledgehammer

## 6 · PLL step 2 · edges

corners solved; arrows show where each edge goes

After any last-layer algorithm, turn U to line the top layer up with its centres.



### Opposite edge swap

Any angle

**M2 U' M2 U2 M2 U' M2**



### Adjacent edge swap

Swapping pairs at front and right

**M' U' M2 U' M2 U' M' U2 M2**



### Three edge, right to left

Solved edge at the back; right edge goes left

**R U' R U R U R U' R' U' R2**

R U' · R U R U take out twice · back in



### Three edge, left to right

Solved edge at the back; left edge goes right

**R2 U R U R' U' R' U' R' U R'**

R2 U · R U R' U' · R' U' R' U R'